

Definition**Positive Definiteness**

Let T be a self-adjoint linear operator on an inner product space V . Then T is called **positive semidefinite** if

$$\langle T(v), v \rangle \geq 0 \quad \text{for any } v \in V.$$

Furthermore, T is called **positive definite** if

$$\langle T(v), v \rangle > 0 \quad \text{for any } v \in V.$$

Note that a positive definite operator is also positive semidefinite, but the converse is not generally true.

Proposition

Suppose V is an inner product space and T is a linear operator on V .

- If T is positive semidefinite, then all the eigenvalues of T are real and nonnegative.
- If T is positive definite, then the eigenvalues are real and strictly positive.

Proof: In either case, since T is self-adjoint, all the eigenvalues of T are real. By the Spectral Theorem, T has at least one eigenvalue.

Let λ be a real eigenvalue of T with a corresponding eigenvector $v \in V$. Then

$$\langle T(v), v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle = \lambda \|v\|^2.$$

If T is positive semidefinite, we know $\langle T(v), v \rangle \geq 0$. Hence $\lambda \|v\|^2 \geq 0$. But since v is nonzero, $\|v\|$ is strictly positive. It follows that λ is nonnegative.

If T is positive definite, then $\lambda \|v\|^2 > 0$. Hence λ is strictly positive. ■

Proposition

Suppose V is an inner product space and T is a positive semidefinite linear operator on V . Then there exists a positive definite operator S such that $S^2 = T$.

Proof: Since T is self-adjoint, by the Spectral Theorem, there is an operator U such that

$$UTU^* = D$$

where D is a diagonal operator and the columns of U are orthonormal. Hence

$$UU^* = I = U^*U$$

meaning $U^* = U^{-1}$.

Since D is a diagonal operator, there is some basis β for V such that

$$[D]_{\beta} = [UTU^{-1}]_{\beta} = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix}$$

where every λ_i is nonnegative. Consider the matrix representation of the operator $\sqrt{D}$ as follows:

$$[\sqrt{D}]_{\beta} = \begin{pmatrix} \sqrt{\lambda_1} & & \\ & \sqrt{\lambda_2} & \\ & & \ddots \\ & & & \sqrt{\lambda_n} \end{pmatrix}$$

Let $S = U^{-1}\sqrt{D}U$. Then

$$\begin{aligned} S^2 &= (U^{-1}\sqrt{D}U)^2 = U^{-1}\sqrt{D}(UU^{-1})\sqrt{D}U \\ &= U^{-1}\sqrt{D}\sqrt{D}U \\ &= U^{-1}DU = T. \end{aligned}$$

Therefore, $S^2 = T$. ■

Exercise

Find the square root of the positive semidefinite matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Definition

An operator U on an inner product space V is **unitary** if

$$UU^* = U^*U = I_V.$$

If V is a real inner product space, then U is called **orthogonal**.

Unitary

Example

Consider the matrix

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Then

$$AA^* = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = A^*A$$

hence A is unitary.

Proposition

A unitary operator U on an inner product space V preserves length; that is,

$$\|U(v)\| = \|v\| \quad \text{for any } v \in V.$$

Proof: Let $v \in V$. Then

$$\begin{aligned}\|U(v)\|^2 &= \langle U(v), U(v) \rangle \\ &= \langle v, U^*(U(v)) \rangle \\ &= \langle v, v \rangle = \|v\|^2.\end{aligned}$$

Hence $\|U(v)\| = \|v\|$. ■

Proposition

The eigenvalues of a unitary operator lie on the unit circle. That is, if λ is an eigenvalue of a unitary operator U on V , then $|\lambda| = 1$.

Proof: Let λ be an eigenvalue of U with a corresponding eigenvector $v \in V$. Then

$$\begin{aligned}\|v\| &= \|U(v)\| = \|\lambda v\| \\ &= \sqrt{\langle \lambda v, \lambda v \rangle} = \sqrt{\lambda^2 \langle v, v \rangle} \\ &= |\lambda| \|v\|\end{aligned}$$

So $|\lambda| \|v\| = \|v\|$, meaning $|\lambda| = 1$. ■

Proposition

Let U be a linear operator on a finite-dimensional inner product space V . Then the following statements are equivalent:

- (i) U preserves inner products, i.e. $\langle U(v), U(w) \rangle = \langle v, w \rangle$ for any $v, w \in V$
- (ii) U is unitary
- (iii) U maps any orthonormal basis for V to another orthonormal basis
- (iv) $\|U(v)\| = \|v\|$ for any $v \in V$

Proof of (i) $\iff$ (ii): Suppose U preserves inner products. Let $v, w \in V$. Then

$$\langle v, U^*(U(w)) \rangle = \langle U(v), U(w) \rangle \quad \text{and} \quad \langle v, U^*(U(w)) \rangle = \langle v, w \rangle.$$

So $U^*(U(w)) = w$ for any $w \in V$, meaning $U^*U = I$. Similarly, $UU^* = I$.

Conversely, suppose U is unitary. Then for any $v, w \in V$,

$$\langle U(v), U(w) \rangle = \langle v, U^*(U(w)) \rangle = \langle v, w \rangle. \quad \text{■}$$